(iii) P (atmost 7 heads)

$$= P(x \leq 7) = 1 - P(x > 7)$$

$$= 1 - \{P(x = 8) + P(x = 9) + P(x = 10)\}$$

$$= 1 - 1 - \left\{\frac{10C_8}{1024} + \frac{10C_9}{1024} + \frac{10C_{10}}{1024}\right\}$$

$$= 1 - \frac{56}{1024}$$

$$= \frac{968}{1024}$$

Example 10.4

With usual notation find p for Binomial random variable X if $n = 6$ and $9\,P(x = 4) = P(x = 2)$

Solution:

$$P(X = x) = nC_x\, p^x\, q^{n-x},\ x = 0,1,2,\ldots n$$

$$X \sim B(6, p) \Rightarrow P(X = x) = 6C_x p^x\, q^{(6-x)}$$

Also $9 \times P(X = 4) = P(X = 2)$

$\Rightarrow$ $9 \times 6C_4 p^4 q^2 = 6C_2\, p^2\, q^4$

$\Rightarrow$ $9\,p^2 = q^2$

$\Rightarrow$ $3p = q$ as $p, q > 0$

$3p = 1 - p$

$4p = 1$

$\Rightarrow$ $p = \frac{1}{4} = 0.25$

Example 10.5

A Binomial distribution has parameters n=5 and p=1/4. Find the Skewness and Kurtosis.

Solution:

Here we are given n=5 and $p = \frac{1}{4}$

Skewness $= \frac{q - p}{\sqrt{npq}}$